

Alejandro Mezio

Quantitative Data Scientist | Predictive Modeling & Machine Learning | PhD in Physics

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Presentation

Data Scientist with a PhD in Physics and a strong background in quantitative modeling, statistics, and scientific computing. Experienced in Python-based data analysis, predictive modeling, and machine learning workflows, developed through academic research, formal postgraduate training, and applied projects.

Proven ability to work with complex datasets, build and evaluate predictive models, and communicate technical results clearly to diverse audiences. Oriented towards applied AI and data science research roles, with a particular interest in ML/AI model evaluation, stochastic processes, and rigorous quantitative analysis.

Skills

Programming and Computing

- Python, Fortran, Git, Bash, Jupyter Notebooks, LaTeX.
- SQL for analytical querying and dataset construction.

Machine Learning and Statistical Modeling

- **End-to-end data science workflows:** data collection, cleaning, exploratory analysis, feature engineering, model training, evaluation, and iterative refinement.
- **Design and articulation of complex problem-solving workflows,** including step-by-step reasoning and clear explanation of assumptions and intermediate steps.
- **Predictive modeling,** with emphasis on error analysis, robustness, and interpretability rather than purely optimized performance.
- **Statistical inference and decision-making:** hypothesis testing, confidence intervals, uncertainty quantification, and model comparison.
- **Exploration of model behavior** under varying assumptions and probabilistic reasoning in systems with incomplete or noisy data.
- **Time-series modeling and forecasting,** incorporating trend, seasonality, and exogenous variables.
- **Unsupervised learning:** clustering, dimensionality reduction, embeddings, and similarity-based reasoning.
- **Foundations of deep learning:** neural networks and CNNs; familiar with transformer-based architectures and self-attention mechanisms.
- **Data visualization and storytelling** to communicate insights and trade-offs clearly to technical and non-technical stakeholders.

Mathematics and Physics

- Strong background in applied mathematics, probability theory, stochastic processes, statistical mechanics, optimization, numerical simulation.
 - Extensive experience modeling complex systems where analytical solutions are not available and simulation-based approaches are required.
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Work Experience

Research Associate

CONICET, Institute of Physics Rosario (IFIR) & National University of Rosario (UNR)

2021 - Present | Rosario, Argentina

- Applied statistical and numerical modeling to analyze complex datasets and evaluate competing quantitative models.
- Designed reproducible computational workflows to ensure robustness, validation, and interpretability of results.
- Translated abstract research questions into tractable quantitative problems, iterating based on empirical results.

Lecturer - Simulation and Stochastic Processes | “Decision-analysis”

Universidad Católica Argentina (UCA)

2026 - Present | Rosario, Argentina

- Teaching Monte Carlo methods, stochastic processes, and simulation techniques applied to real-world data science problems.
- Teaching Decision Analysis, with emphasis on probabilistic reasoning, uncertainty quantification, and evaluation of decisions under incomplete information.
- Guiding students in probabilistic modeling and uncertainty-aware reasoning.

Teacher in Charge of Assignments – Physics | Mathematics

Physics Department, National University of Rosario (UNR)

2014 - Present | Rosario, Argentina

- Designed and delivered analytical and computational exercises focused on numerical methods and modeling.
- Specialized in breaking down complex quantitative concepts into clear, structured, step-by-step explanations for diverse audiences.

Postdoctoral Research Fellow - IFIR & The University of Queensland

2014 - 2021 | Argentina & Australia

- Developed and optimized computational models for complex physical systems, including large-scale numerical simulations.
- Applied stochastic and numerical methods to study systems with non-trivial interactions and emergent behavior.
- Collaborated in interdisciplinary teams, integrating modeling, statistical analysis, and computational techniques.
- Mentored junior researchers and presented results through peer-reviewed publications and international conferences.

Applied Data Science Projects

Delivery Time Prediction for E-Commerce Logistics (Olist Dataset)

GitHub: github.com/alemezio/Ecommerce_Brazil_project

End-to-end regression on 90K orders from 9 relational tables (~1M geolocation records): per-table data audits and domain-driven feature engineering (carrier chargeable-weight rule, seller-customer distance).

- Compared 9 model families under a seeded, pipeline-encapsulated CV protocol; XGBoost cut MAE 22% vs. the strongest trivial baseline and 9% vs. linear regression (test MAE 4.38 days), and a P90 quantile model delivering promise dates ~4 days tighter than the platform's at calibrated 90% coverage.

- Key contribution: evaluated the same model under three protocols. A deployment-faithful temporal split confirmed the point estimate (forward MAE 3.82 on the final window).
- Random CV understated deployment uncertainty by ~25x relative to the temporal split (4.84 ± 1.16 MAE across periods); seller-grouped CV bounded entity memorization at 2.9%.
- Production packaging: src/ pipeline reproducing the notebook checkpoint to the last decimal, train/predict CLI with unseen-category guards, and smoke tests running on a committed data sample.
- Tools: Python, pandas, scikit-learn, XGBoost, SHAP, geopy, pytest.

Absenteeism Prediction: Leakage-Aware ML Case Study

GitHub: github.com/alemezio/absenteeism-prediction

- Built a binary classifier to predict excessive employee absenteeism (>3 hours) from 700 medical-leave records spanning 34 employees (UCI Absenteeism at Work dataset).
- Detected person-identity leakage in the standard random-split protocol and rebuilt evaluation with group-aware splits (GroupShuffleSplit, StratifiedGroupKFold); showed that the honest protocol reverses model rankings: high-capacity ensembles memorize employee identity (train AUC ~1.0) while regularized logistic regression generalizes.
- Reported ROC-AUC 0.73 ± 0.09 on unseen employees (group-aware 5-fold CV), treating the 0.82 held-out test result as confirming evidence rather than the headline metric.
- Delivered decision-ready analysis beyond accuracy: threshold selection under explicit business cost ratios, probability calibration, misclassification cost by absence severity, coefficient stability checks, and an ethics assessment for HR deployment (protected categories).
- Shipped a production-style repository (src/ module, prediction CLI, smoke tests enforcing notebook-script consistency; MIT licensed, reproducible from a fresh clone), extending a course case study well beyond its original scope.
- Tools: Python, pandas, scikit-learn.

Education

Postgraduate Diploma in Data Science, Artificial Intelligence, and their Applications in Economics and Business

2025 | National University of Córdoba (UNC), Argentina

Ph.D. in Physics

2009 - 2014 | National University of Rosario (UNR), Argentina

Thesis: Quantum Spin Systems and Frustrated Magnetism

Licenciado en Física (equivalent to Master's Degree in Physics)

2009 | National University of Rosario (UNR), Argentina

Graduated first in class (highest GPA)

Certifications and Training (Data Science & AI)

- Machine Learning for Aspiring Data Scientist - Udemy (2024)
- Complete Data Science Bootcamp 2024 - 365 Careers (2024)
- Python & Django Framework - CoderHouse (2023)

Languages

Spanish (native), English (fluent), French/Italian (basic reading)